PP3 reference card

http://pp3.sourceforge.net

for version 1.3

General structure

```
# Some introductory comments,
# i.e. what the file is about to do.

[Section I: Parameters.]

objects_and_labels

[Section II: Delete/add/modify objects and/or labels]
```

Both sections may be empty. If section II is empty, **objects_and_labels** should be omitted.

Section I: Parameters

The following top level keywords are allowed here: **filename**, **set**, **switch**, **color**, **line_width**, **line_style**, and **penalties**[1].

```
filename output orion.tex
```

writes the result to the file `orion.tex`. The file extension must be `.tex` and must not be omitted.

```
filename stars               stars.dat
filename nebulae             nebulae.dat
filename label_dimensions    labeldimens.dat
filename constellation_lines lines.dat
filename boundaries          boundaries.dat
filename milky_way           milkyway.dat
```

This lets PP3 look for the respective data in the given file. The filenames must be full paths from the current directory.

[1] not explained on this reference card

```
filename latex_preamble mypreamble.tex
```

makes PP3 include the file `mypreamble.tex` with e. g. font parameters in every LaTeX run.

```
filename include myproject.pp3
```

includes `myproject.pp3` that itself has a normal input script structure, except for the limitation that it mustn't include yet another file.

```
set center_rectascension  5.8
set center_declination    0
```

This sets the center of the printed map to the celestial coordinates $(5^{\mathrm{h}}48', 0°)$.

```
set box_width  15
set box_height 15
```

sets the dimensions of the printed map in centimetres. (15 for both is the default.)

```
set grad_per_cm 4
```

sets the scale of the map in degrees per centimetre. (4 is the default.)

```
set constellation ORI
```

tells PP3 to highlight the boundaries of constellation Orion. Must be in uppercase.

```
set shortest_constellation_line 0.1
```

makes PP3 suppress all constellation lines that are shorter than 0.1 centimetres. (This is the default.)

```
set label_skip 0.06
```

sets the distance between the outer rim of the celestial object and its label to 0.06 centimetres. (This is the default.)

```
set minimal_nebula_radius 0.1
```

means that all nebulae that would be smaller than 0.1 centimetres (in radius) are printed with a radius of *exactly* 0.1 cm. (This is the default.)

```
set faintest_cluster_magnitude 4.0
```

suppresses by default all stellar clusters that are fainter than 4.0^{m}. (This is the default.)

```
set faintest_diffuse_nebula_magnitude 8.0
```

suppresses by default all emission and reflection nebulae and all galaxies that are fainter than 8.0^{m}. (This is the default.)

```
set faintest_star_magnitude 7.0
```

suppresses by default all stars and all galaxies that are fainter than 7.0^{m}. (This is the default.)

```
set faintest_star_with_label_magnitude 3.7
```

means that for stars as bright as 3.7^{m} or brighter get automatically generated labels. (This is the default.)

```
set faintest_star_disk_magnitude 4.5
```

means that all stars that are fainter than 4.5^{m} are printed as mere dots instead of small circles. (This is the default.)

```
set minimal_star_radius 0.015
```

This sets the radius of the dots for all stars fainter than the limit determined by `faintest_star_disk_magnitude` to 0.015 centimetres. (This is the default.)

```
set star_scaling 2.0
```

makes all star circles two times bigger than normal. (Default is 1.0.)

```
set fontsize 12
```

sets the global font size to 12 pt. (Default is 10.)

```
switch milky_way           off
switch nebulae             on
switch grid                on
switch ecliptic            on
switch boundaries          on
switch constellation_lines on
switch labels              on
```

This switches the display of the Milky Way off, and the nebulae, coordinate grid, ecliptic, constellation boundaries, boundaries of the highlighted constellation, and all labels on. This is the default.

```
switch eps_output on
switch pdf_output on
```

The first line makes PP3 generating an EPS file from the map via a dvips call, the second creates a PDF file additionally by calling ps2pdf.

```
switch colored_stars off
```

This disables PP3's default behaviour to try to illustrate the star's actual colour in the map. Instead, all stars get the same colour explicitly defined by "**color** stars".

```
color background              0   0    0.4
color grid                    0   0.298 0.447
color ecliptic                1   0    0
color boundaries              0.5 0.5  0
color highlighted_boundaries  1   1    0
color constellation_lines     0   1    0
color milky_way               0   0    1
color nebulae                 1   1    1
```

This sets the colours of these map elements to the respective RGB colour. The colour values range from 0 to 1. What you see here are the default values.

```
color stars                   1   1    1
```

This sets the colour of stars to white (the default), however only if you switch `colored_stars` off.

```
color labels                  0   1    1
```

sets the colour of all automated labels to cyan (the default).

```
color text_labels             1   1    0
```

sets the colour of all user defined labels to yellow (the default).

```
line_width grid                    0.025
line_width ecliptic                0.018
line_width boundaries              0.035
line_width highlighted_boundaries  0.035
line_width nebulae                 0.018
line_width constellation_lines     0.035
```

This sets the line widths of these map elements to the respective width in centimetres. What you see here are the default values.

```
line_style grid                    solid
line_style ecliptic                dashed
line_style boundaries              dashed
line_style highlighted_boundaries  dashed
line_style nebulae                 solid
line_style constellation_lines     solid
```

This sets the line styles of these map elements to the respective style. What you see here are the default values.

Section II: Objects and labels

The following top level keywords are allowed here: **add**, **delete**, **reposition**, **add_labels**, **delete_labels**, **set_label_text**, and **text**.

```
add IC 5070  NGC 7000;
```

adds the nebulae IC 5070 and NGC 7000 to the map with their labels.

```
delete LEO 63  HD 97605 ;
```

removes the two stars 63 Leo and HD 97605 from the map.

```
reposition LEO 32 SE ;
```

positions the label of 32 Leo to the lower right (SE = **S**outh **E**ast).

```
add_labels M 66  LEO 24 ;
```

adds the labels of M 66 and 24 Leo to the map. The positioning is still automatic.

```
delete_labels M 43 ;
```

deletes the label of the nebula M 43 from the map, *not* the nebula itself.

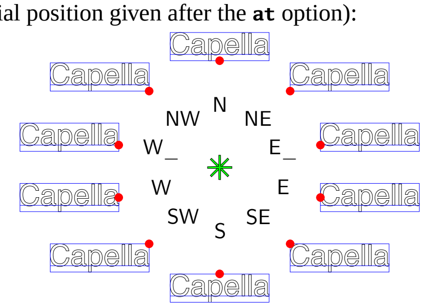
```
set_label_text UMA 50 "\\footnotesize Dubhe"
```

replaces the default label text for 50 UMa with "`\footnotesize Dubhe`".

```
text "\\small Wolf 359\\hskip0.3em
      \\psdots[dotstyle=+,dotangle=45](0,0)"
     at 10.902 7.32
     color 0.3 0.3 0.9333 towards NW ;
```

This prints the LaTeX commands between the "`...`" at the celestial point $(10^h 54^m 7.2^s, 7.32°)$, in a grey-blue colour, and to the upper left of the point.

This graphics shows the possible values for **reposition** and the **towards** option (the large dots lie exactly on the celestial position given after the **at** option):




```
text "$#3$" at 0 20 along declination
            tics rectascension 1 towards N ;
text "$#5$" at 11 0 along declination
            tics declination 10 towards S ;
```

This creates automatically generated labels for the 20° declination circle (every whole hour), and for the 11^h rectascension line (every 10 declination degrees). The following placeholders are available:

- `#1` Rectascension in hours.
- `#2` Declination in degrees.
- `#3` Rectascension in integer hours (truncated, not rounded).
- `#4` Rectascension fraction of hour in minutes (truncated, not rounded).
- `#5` Declination in rounded integer degrees.

LaTeX preamble

```
\usepackage{amsmath}
\usepackage[T1]{fontenc}
\usepackage{eulervm}
\renewcommand*{\sfdefault}{phv}
\AtBeginDocument{\sffamily\boldmath}
```

This sets some global font parameters: We use a sans serif font (Helvetica) as the standard, and Euler for all Greek letters.

```
\usepackage{relsize}
\renewcommand*{\Messier}[1]{%
  \footnotesize{\smaller M}\,#1}
\renewcommand*{\NGC}[1]{%
  \footnotesize{\smaller NGC}\,#1}
\renewcommand*{\IC}[1]{%
  \footnotesize{\smaller IC}\,#1}
```

This resets the catalogue hooks. `\Messier`, `\NGC`, and `\IC` take one parameter which is the catalogue number. In this example they print the catalogue abbreviation a little bit smaller.

```
\renewcommand*{\Starname}[1]{#1}
\renewcommand*{\Label}[1]{#1} % automatic labels
\renewcommand*{\TextLabel}[1]{#1}  % user labels
\renewcommand*{\FlexLabel}[1]{{\bfseries #1}}
\renewcommand*{\TicMark}[1]{{\mdseries
    \scriptsize\mathversion{normal} #1}}
\renewcommand{\DP}{.}
```

Further hooks: This makes all flexes bold and all tics small and non-bold. `\DP` is the decimal point (the dot is the default). In the first three lines you see default definitions: The parameter is just mapped to the output.